

Introduction

Spiking Neural Networks (SNNs) utilize biologically-inspired spikes for energy-efficient communication and computation.

Distributed Learning trains multiple models and then merge into one model.

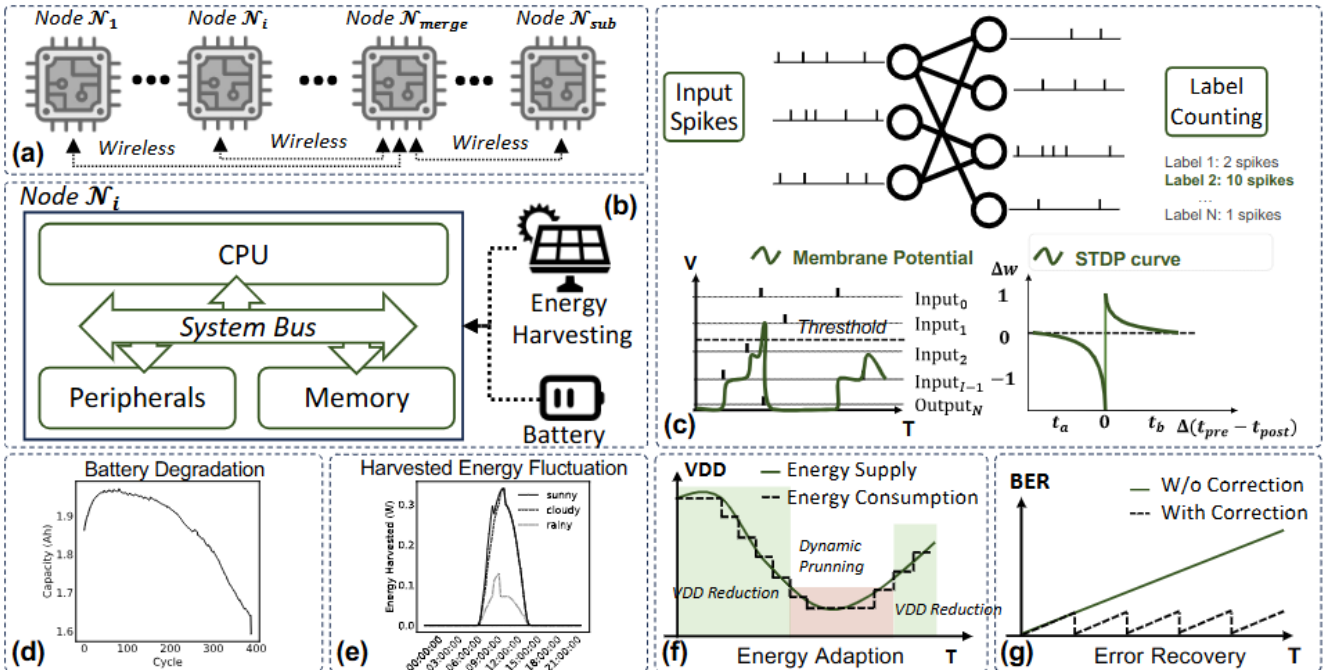
Carbon Neutral Computing Covers SNN energy consumption by harvested energy, such as solar energy.

Challenges:

- The amount of energy harvested fluctuates by weather etc.
- Reducing model parameters leads to performance loss.

Objective: Achieving carbon neutrality in each node of distributed learning while minimizing performance loss.

System Architecture



Energy Optimization Through DRAM

Weather	Sunny			Cloudy			Rainy		
Battery	Fresh	Mid	Wear-out	Fresh	Mid	Wear-out	Fresh	Mid	Wear-out
Energy Deficit (System)	0%	11.45%	27.08%	7.92%	11.45%	27.08%	88.13%	88.13%	88.13%
Energy Deficit (DRAM)	0%	22.90%	54.16%	15.84%	22.90%	54.16%	not feasible	not feasible	not feasible
Power Constraint (Normalized)	1.00	0.78	0.46	0.84	0.78	0.46	—	—	—
Generated Voltage	1.10	0.96	not feasible	1.00	0.96	not feasible	—	—	—
Predicted Accuracy	0.9186	0.9181	—	0.9185	0.9181	—	—	—	—
Actual Accuracy	0.9183	0.9194	—	0.9210	0.9194	—	—	—	—

Energy Optimization Through Dynamic Pruning

S_G	E_{target} (Wh, %)	Sampled ($N_{tim}, T_{timestep}$)	Generated ($N_{tim}, T_{timestep}$)	Accuracy for Sampled ($N_{tim}, T_{timestep}$)	Accuracy for Generated ($N_{tim}, T_{timestep}$)	$E_{inference}$ for Generated ($N_{tim}, T_{timestep}$) (Wh)
700	9.42 (100.00%)	(700, 250)	(700, 250)	0.8709	0.8709	9.42
	6.86 (72.82%)	(590, 250)	(659, 237)	0.8674	0.8699	7.10
	4.26 (45.23%)	(350, 250)	(569, 177)	0.8519	0.8682	3.60
	2.12 (22.51%)	(350, 125)	(458, 104)	0.8531	0.8628	1.90
	1.84 (19.54%)	(175, 125)	(439, 92)	0.8281	0.8573	1.70
	1.40 (14.87%)	(100, 100)	(363, 80)	0.7921	0.8551	1.40
	1.10 (11.68%)	(50, 80)	(50, 80)	0.7828	0.7828	1.10

References

- [1] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers in computational neuroscience*, vol. 9, p. 99, 2015.
- [2] Hanyu Yuga and Khanh N. Dang, "EnsembleSTDP: Distributed in-situ Spike Timing Dependent Plasticity Learning in Spiking Neural Networks", 17th IEEE MCSoc 2024 (accepted for publication).